WEB APPLICATION ARCHITECTURE

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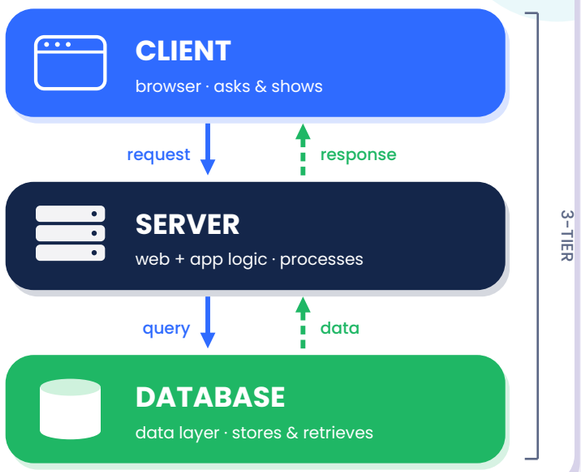
"What happens behind the screen when a user interacts with a web application?"

1 What is Web App Architecture?

The blueprint of how a web app is organised: the browser, the server, the database and the security that links them. Together they turn one click into a result on your screen.

Client · Server · Database · Security

2 Client–Server Architecture



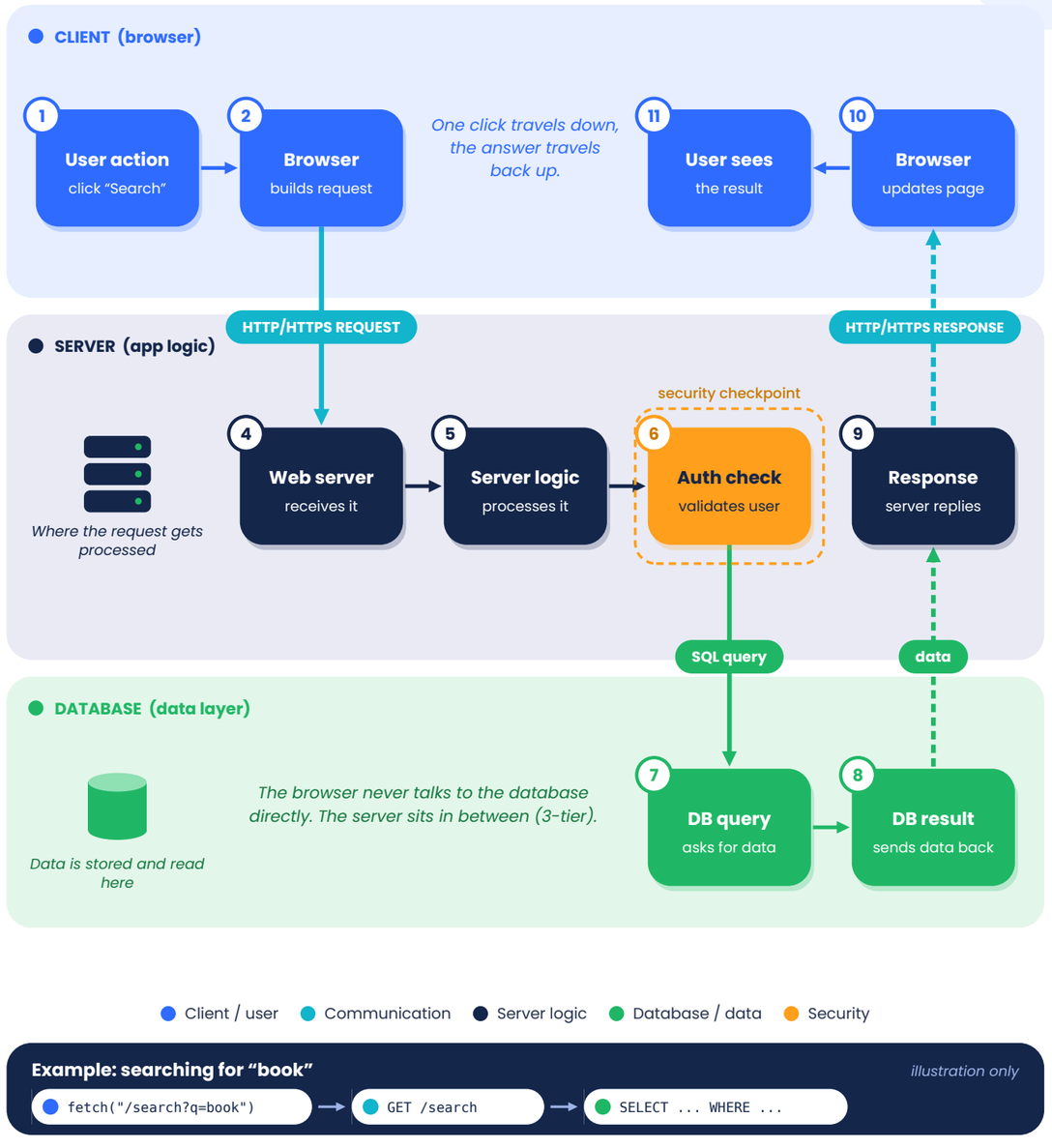
3 HTTP vs HTTPS

HTTP
The rules for sending requests and responses

HTTPS
HTTP + TLS encryption, so data in transit is protected

HTTPS protects data in transit only. Apps still need validation, login and safe sessions.

7 Complete Web Application Flow



4 Request & Response

REQUEST *browser to server*

```
GET /search?q=book HTTP/1.1
```

Method what to do: GET, POST
URL where to go + query
Headers cookies, browser, type
Body data sent (forms)

RESPONSE *server to browser*

```
HTTP/1.1 200 OK
```

Status the result code
Headers type, cookies
Body HTML, JSON, images

200 OK · 301 Moved · 404 Not found · 500 Error

GET read · POST send · PUT update · DELETE remove

5 Workflow & Real World

BUILD WORKFLOW

Plan → Design UI → Build client → Events → Server logic → Database → Security → Test → Deploy

APP LIFE CYCLE

Design → Development → Testing → Deployment → Maintenance

SAME ARCHITECTURE, DIFFERENT APPS

Banking · E-commerce · College portal · Healthcare · Social media · Learning · Booking

6 Database Connectivity

App → Connection → Query → Result

id	task
101	view creation
102	view cursor

```
CREATE VIEW v101 AS
SELECT name FROM Students;
DECLARE c102 CURSOR FOR
SELECT * FROM v101;
```

SQL talks to relational databases

8 Conclusion

A web app is teamwork: the browser asks, the server thinks, the database stores and security protects every step. That is what happens behind every click.

CLIENT shows the page and sends requests

SERVER checks, runs logic and replies

DATABASE stores data and returns it

SECURITY HTTPS, login and sessions protect all

JOURNEY OF A CLICK: USER → REQUEST → PROCESS → DATA → SECURITY → RESPONSE → USER